

Table 3. Model and stability audit

Domain	Value
Discovery cohort	6,048 adults
Selected configuration	SVD 75 requested; UMAP 10 components, 50 neighbors; HDBSCAN n
Final solution	3 non-noise phenotypes plus algorithmic noise/outlier group
Mean pairwise ARI	0.9935
Mean pairwise NMI	0.9885
Mean noise rate	7.10%
Most destabilizing ablation	Remove lipids (mean ARI vs full objective reference 0.263)
Temporal profile matching	Best-match profile correlations 0.714 to 0.909